3.2.2 Materials

- ### Bulk properties of solids

Density $\rho = \frac{m}{V}$

Hooke's law, elastic limit, experimental investigations.
$F = k\Delta L$
Tensile strain and tensile stress.
Elastic strain energy, breaking stress.
Derivation of $energy\ stored = \frac{1}{2} F\,\Delta L$
Description of plastic behaviour, fracture and brittleness; interpretation of simple stress-strain curves.

- ### The Young modulus

$$\text{The Young modulus} = \frac{tensile\ stress}{tensile\ strain} = \frac{FL}{A\ \Delta L}$$

One simple method of measurement.
Use of stress-strain graphs to find the Young modulus.